

Greening Residential Maintenance



Jim LaRue, a green building consultant, has prepared a green building series for the city of Cleveland, Ohio.

JIM LARUE

This is the final installment of my series on green residential building. In this column, I want to emphasize the importance of residential behavior on green buildings. There are a number of actions and behaviors that should be kept in mind if the residents are to achieve the full measure of what a greenbuilt home can be for them. Their actions are the final determiner of the home's value and homeowners deserve to know as much as they can to make it work best for them.

Ensuring High Indoor Environmental Quality

One easy step residents can take is to protect and clean all interior surfaces with non-toxic cleaners. This includes floors, countertops, furnishings, and windows. There are a number of environmental reasons for using non-toxic cleaners, not the least of which is the health of residents. The chemical composition of many cleaning solutions is being questioned, and with it, healthier options are being created. Organizations such as Environmental Health Watch and the National Center for Healthy Housing provide information on some of these options.

All equipment in your green home should also be properly used and maintained. Filters should be regularly replaced on heating and cooling equipment. Thermostats and humidistats should be adjusted so they are operating at their most effective temperatures and humidities.

A failure to do so can result in additional energy costs, damage to the mechanical equipment, and poor interior environmental conditions, including mold growth.

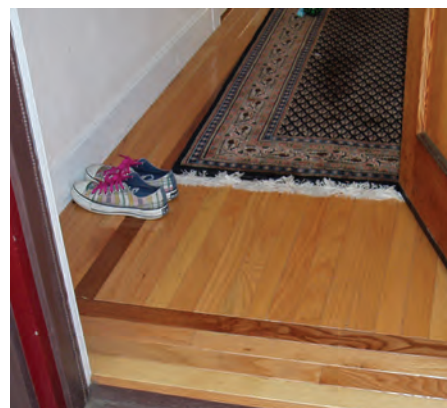
Bath and kitchen fans should also be run at regular intervals, since when they are not used regularly, this lack of circulation can result in increased moisture levels that can feed mold growth. Input fans or heat recovery ventilators (HRVs) need to be regularly maintained so they are not bringing in excess moisture or air at too low or too high temperatures. Water heating equipment, including sump pumps and other plumbing devices, need to be regularly cleaned and/or snaked to reduce water backup problems and increase the life of the equipment.

Since green homeowners have put much time and effort into making sure that their living space is green and healthy, it doesn't make sense to buy objects and products for the home that are going to compromise the air quality and living quality of the home. Instead, bring furniture and materials into living spaces that have low-or non-toxic components. While many manufacturers of furniture and carpeting are moving to reduce the health risks from their products, the requirement for fire retardants in such products is a significant risk for consumers.

Environmental Building News advises asking product manufacturers about halogenated flame retardants in their products and suggests avoiding products with combustible materials as much as possible with particular attention to foam insulations and padding in furniture and under carpeting. Many cabinets and furniture pieces may contain urea formaldehyde, which is a binding agent used to hold particleboard together. It releases gases when the area in which it is located is warm and moist (as in a kitchen or



OZAN WILLIAMS



ELKA KARL

(top) The chemical composition of many cleaning solutions is being questioned, and with it, healthier options are being created.

(bottom) It is essential to limit foot traffic in the home through the use of walk-off mats just outside entry doors and inside those doors, combined with removal of shoes after residents are inside.

bath) and these gases can cause some respiratory distress and are on a risk list for possible cancers.

Another way to ensure your green home's quality is by controlling hazards in and around the home. Make sure all equipment that uses natural gas for heat are operating efficiently and do not backdraft when competing with ventilation equipment. Testing for and reducing radon levels in homes is also essential. In our new green homes we have installed venting systems under the basement slab that can be vented to the exterior in the event that the home tests positive for radon.

The most obvious hazard to indoor environmental quality (IEQ) is finally beginning to be taken seriously by

green building



ELKA KARTL

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residents: tobacco smoke. In our healthy house work we have been able to persuade heavy smokers to at least do it outside the living space if they cannot seem to quit completely.

In addition, it is essential to limit foot traffic in the home through the use of walk-off mats just outside entry doors and inside those doors, combined with removal of shoes after residents are inside. By using this simple practice, homeowners can dramatically reduce the input of such things as lead dust (most common in inner-cities) or herbicides and pesticides (most common in suburban communities) into their homes.

All homes are likely to have an outbreak of some pest at some point in the life of the home, whether it's fleas, rats, or ants. The best way to control pests in living spaces is through using Integrated Pest Management (IPM). It is no longer necessary to use toxic sprays to kill insects. IPM provides a number of strategies for reducing these pests, including changing the conditions that may draw pests in the first place, then, when necessary, uses baits strategically placed that cause the pests to take the chemicals they ingest back to the nests where it can have an impact on the remainder of the colony. Eliminating the root causes and pest harborages means far less chemicals used that can be harmful to residents and the environment.

The importance of effective storage strategies should also not be underestimated. I have investigated

hundreds of homes with wet basements over the years and found all manner of cellulose products on the floors and against walls. This creates lots of mold, which in turn compromises the home's IEQ. Installing simple plastic or metal shelf systems to hold such products can solve the problem and reduce a potential health hazard.

By controlling pet dander, especially cat dander, homeowners can also improve their home's IEQ, since pet dander can be a significant asthma trigger. Controlling all the potential by-products of pets around the house can determine whether a residence is operating in a healthy fashion.

While most people may think that plants are a boon to IEQ, this is in fact not the case. I have investigated many homes with severe mold problems that were significantly influenced by an overabundance of plants. The water required to maintain the plants also dramatically affected the interior humidity that fostered mold growth.

Walking the Talk

Recycling is an integral part of what it means to build green, not just in the building process, but in the long term maintenance of a home. For that reason, residents should strive to recycle as much waste as possible. Reducing the volume of what we use and carefully recycling the by-products of what we use is consistent with all that green building represents. With improvements in recycling programs across the country, more and more communities and counties are making it easier to do so. We left local community recycling containers in each of our green homes. Our county regularly schedules days when you can bring old paint, gasoline, oil, and other such liquids and other days when you can bring electronic equipment that is no longer being used.

Go the next step with your recycling practices by composting organic waste. While composting was once considered limited to rural homes and communities, urban communities are increasingly permitting this activity. Some apartment and condo owners

find that worm bins, which can be placed under their sinks, are an effective composting method in small space living. Additionally, many cities are allowing kitchen scraps—including take-out paper containers and pizza boxes!—to be placed in the green bins that traditionally only accepted landscaping waste.

If your green home is lucky enough to include a spacious outdoor area, strive to control pests in exterior areas by only using organic fertilizers, herbicides, and pesticides. Control ice buildup with organic substances as well. Both of these measures will help ensure that your yard is safe for pets, children, and grown-ups as well.

If at all possible, avoid use of gas combustion yard equipment. These small gas units generate an enormous amount of pollutants during their combustion. The use of battery-charged equipment is superior and the use of electrical power for such equipment is preferable to gas fired. Human-powered yard equipment, such as push mowers and hand-operated tree trimmers and saws, are the best option yet.

A few years ago, a Canadian study was done to determine how much impact residents have on the residence in which they live. Six identical homes were constructed and six average middle class families moved into them. Within six months each of the homes had generated significant pollution levels from the activities of the families. The thoughtfulness that goes into creating a green home usually means the residents are more likely to pay attention to how they maintain their home. Should they not, the integrity of a green home is significantly compromised at a cost in dollars, reduced durability, and diminished health. Greening home maintenance is therefore essential!!

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For more information:

For more information on maintaining a healthy home, find tips at Environmental Health Watch (www.ehw.org) or the National Center for Healthy Housing (www.centerforhealthyhousing.org).